

CONNECT360: AN INTELLIGENT AND SECURE PLATFORM FOR SKILLED LABOUR BOOKING

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Abstract - Finding trusted skilled service workers remains a significant problem in the informal home-services space. Customers are usually reliant on private referrals or manual searches to find suitable workers for a specific service that meet their skills, availability, experience, trust, and distance requirements. Skilled workers, on the other hand, have few opportunities for digital exposure of their abilities and do not have access to a robust platform to handle requests and build credibility. The situation becomes even more challenging for urgent service requests, requiring identifying an appropriate worker within a short period.

This paper proposes the Connect360 smart worker matching algorithm and a context-aware platform that offers skilled worker discovery, worker verification, service finding, Scheduled Booking, and Priority Booking. Connect360 CSWMA (Connect360 Smart Worker Matching Algorithm) ranks potential candidates in accordance with different factors with varied weights depending on the request type, including Scheduled Booking and Priority Booking. The proposed approach introduces automatic rematching for Priority Booking and road-based worker tracking alongside a role-aware conversational assistant for service booking. In addition, the paper implements a system that secures

worker verification documents and offers role-based access control. The proposed solution is evaluated using AWS serverless infrastructure to measure matching latency, ranking sensitivity, transactional concurrency, spatial routing, and PII redaction experiments. The results demonstrate the ability of the context-aware platform to enable worker discovery and booking in a secure serverless home-service environment.

Key words— Skilled service workers, home services, worker matching, context-aware recommendation, Priority Booking, worker verification, serverless computing, artificial intelligence, CSWMA.

I. INTRODUCTION

The proliferation of digital service platforms has transformed the ways in which customers search and access services. Nevertheless, the problem of finding an appropriate skilled service worker for home-service needs such as electrical, plumbing, carpentry, cleaning, or equipment maintenance still persists. Unlike products, the process of choosing a service worker requires considering a number of factors. For example, in addition to the service scope, a customer may want to take into account the worker's skills, availability, experience, reliability, reputation, verification status,

and proximity. Since a worker may be available but located far from the customer's residence or provide high-quality service but be unwilling to work at a specific time, choosing the most suitable worker based on multiple criteria is critical. As can be seen, the task of picking up a home-service worker based on the provided service description and personal preferences involves a complex multifactorial decision-making process.

As the informal home-services market poses a significant challenge to both customers and service workers, it is necessary to address it through a comprehensive solution. In the current situation, where customers resort to word-of-mouth, local networks, and Google search to find a worker, a multifactorial recommendation system would help sort service providers according to customers' individual preferences. At the same time, since informal workers do not have a strong digital presence, a recommendation system would allow skilled workers to advertise their services and attract clients by showcasing their qualifications, working conditions, and other characteristics. Overall, the abovementioned considerations contribute to the necessity of building such a system, which would enable both buyers and sellers of home-services in local networks to communicate, share information, and conduct transactions more efficiently.

The problem becomes more complicated when service requirements are different in terms of urgency. For a Scheduled Booking, the customers will have enough time to evaluate the suitability, trust, experience, availability, reliability, and distance of a worker before making a decision. However, for a Priority Booking, the allocation process should be more time-critical since the worker's availability and proximity to the service location are significant factors. Therefore, a worker may be suitable for a certain planned service, but at the same time, he may be unavailable or located too far from the customer's location to provide an immediate service. Thus, a home-service marketplace should consider different booking contexts to maintain service suitability and trust.

This paper proposes Connect360, the serverless home-services marketplace, which connects customers with skilled service workers through a central platform. It offers role-based workflows for customers, workers, and administrators. With a customer's interface, one can search for the trusted workers, check their skills, experience, availability, and perform a Scheduled or Priority Booking. Using the worker's interface, they can manage his professional account, check customers' requests, accept or deny booking requests, update booking status, and upload documents for verification. Administrators are responsible for managing workers and service categories, reviewing workers' documents,

and storing booking-related information. Connect360 uses AWS serverless infrastructure to ensure the safety of customer data and provide a solid foundation for service management.

In this regard, the Connect360 platform uses a Connect360 Smart Worker Matching Algorithm (CSWMA) to rank the appropriate worker using various attributes related to service and worker. Since scheduled services and priority services have unique needs, different weighing techniques are used for Scheduled and Priority Booking so that the process of ranking can take into account the unique needs of these two types of services. Apart from worker matching, Connect360 also offers worker verification, auto rematching for Priority Booking, booking control to prevent double booking, and road-based live tracking for active Priority Bookings. It uses a role-aware conversational assistant based on Google Gemini to give service interaction that is structured and multilingual, with rule-based fallback. Furthermore, the platform handles the process of worker verification securely using private storage and access control for the documents along with role-based authentication and authorization.

For customer to worker communication, the existing implementation provides calling via device's native dialer from active bookings.

This paper's significant contributions include a serverless skilled home-service platform involving customers, skilled workers, and administrators; the Connect360 Smart Worker Matching Algorithm (CSWMA) that ranks workers on the basis of contextual information; different weighting schemes used in Scheduled and Priority Booking modes; Priority Booking transactions where server-side availability validation, slot locking, and re-matching is performed to minimize booking conflicts and address cases where workers reject or are unavailable for the booking; and an evaluation framework that measures the performance in terms of matching time delay, ranking sensitivity, transactional concurrency, routing, and PII redaction.

II. LITERATURE REVIEW

The related literature falls under local service-provider recommendation, service recommendation, skill-based matching, spatial worker allocation, trust-aware recommendation, artificial intelligence, and cloud-based service systems. Each of the research directions focuses on distinct aspects of recommending appropriate service providers to users. The literature considers the service requirements, worker capabilities, user preferences, trust, location, and task constraints, providing the foundation for developing a skilled home-service booking platform. To find an appropriate local service provider, customers often need to assess the capabilities and reliability of the workers before they

request services. Ezhilnilavan et al. [1] propose a machine learning local community service-provider system that recommends local providers while considering providers' information, verification, preferences, and reviews. Munir et al. [13] examine the developer recommendation and team formation for collaborative crowdsourcing platforms by encoding the developers' skills and task requirements to perform the matching. Kumar et al. [19] suggest a skill-based rule matching approach to job recommendation by considering the qualification and skills of the workers. The approaches highlight the importance of representing the capabilities of the service providers and matching them with the users' requirements. However, skilled home-service recommendation needs to consider worker availability, trust, location, booking requirements, and service urgency.

The service recommendation literature increasingly focuses on user preferences and utilizes multiple sources of information for accurate recommendations. Sun et al. [2] propose a weighted preference optimization approach based on knowledge graphs and large language models for personalized service recommendation. Ge et al. [6] develop a new approach called LLMGCL that utilizes graph contrastive learning and large language models for recommendation. Zhao et al. [14] explore the recommender systems in the age of large language models, whereas Zheng et al. [15] investigate the collaborative semantics and large language models for recommendation. The approaches highlight the potential of using semantic information and language models for capturing structured data and user preferences for recommendation. However, a skilled home-service request requires that the capabilities of the selected worker satisfy the customer requirements and service conditions.

When choosing a service provider, customers often seek trustworthy and reliable workers. Khelloufi et al. [11] propose a hybrid feature and trust aggregation recommender system that incorporates trust information into the recommendation process. Chen et al. [12] examine reliable personalized recommendation with uncertain missing modalities, addressing the challenge of making accurate recommendations using available information. The approaches highlight the importance of trustworthy information for making accurate recommendations. In the context of home service, worker trust is critical and should be considered during recommendation, worker verification, and previous booking records.

When allocating workers to tasks that require traveling to a customer's location, the spatial aspect of the task adds complexity to the service provision process. Zhao et al. [3] propose heterogeneous graph reinforcement learning for dependency-aware multi-task allocation in spatial crowdsourcing by modeling the relationship

between workers and tasks. Wu et al. [5] investigate adaptive task assignment through a human-in-the-loop approach. Deng et al. [7] explore the task recommendation in spatial crowdsourcing by considering the balance between diversity and coverage. Wu et al. [8] develop a prediction-aware adaptive task assignment approach, whereas Zhao et al. [9] propose a geo-partition framework for spatial crowdsourcing task allocation. The approaches highlight the importance of incorporating spatial information and worker-task relationships into the task allocation process. However, home service allocation requires considering these factors in addition to the professional skills, worker trust, experience, and availability to ensure effective service delivery.

In addition, the time factor often plays an essential role in task allocation. Ma et al. [10] propose a clustering-based priority queue approach to spatial task assignment by considering task priorities. Chen et al. [4] examine the fairness-aware task recommendation and allocation, establishing the connection between the recommendation and allocation processes. The approaches highlight the importance of task priority and fairness in task allocation. For home service, the time factor determines the urgency of the service request. As such, it is essential to adjust the worker matching strategy according to the booking requirements. For example, the proximity of the worker to the customer's location is more critical for a Priority Booking than a Scheduled Booking.

Lastly, artificial intelligence can be utilized to develop an intelligent system that interacts with customers and allocates workers to service requests. The approaches mentioned above demonstrate the potential of using large language models and graph neural networks to capture semantic information and improve recommendation accuracy. The local service-provider literature [1] also demonstrates how artificial intelligence can be utilized to discover local workers. However, the focus of the approaches is on the recommendation or artificial intelligence aspects of the service system, thus limiting their application in developing a skilled home-service booking platform that involves customer preference understanding, worker ranking, booking, and service execution.

The cloud-based service literature provides insights into developing the computational infrastructure that supports the service system. Refonaa et al. [16] examine the serverless infrastructure for cloud automation, whereas Ekwe-Ekwe and Amos [17] compare public cloud Function-as-a-Service providers. Adeppady et al. [18] study the dynamic management of cloud computing resources with constrained serverless platforms. Pasam et al. [20] investigate scalable AI pipeline development using cloud-native technologies for building intelligent transactional systems. The

approaches highlight the potential of utilizing cloud-native and serverless technologies for developing distributed and scalable service systems. However, the focus of the literature is on the technical aspects of serverless infrastructures and does not address the end-to-end skilled home-service discovery, booking, and execution processes.

In summary, the literature provides the foundation for developing a skilled home-service booking platform by addressing local provider discovery, skill-based job recommendation, personalized recommendation, trust, spatial allocation, time-based allocation, artificial intelligence, and cloud-based service systems. However, each of the research directions primarily focuses on a distinct aspect of the skilled home-service platform, thus leaving the comprehensive approach to skilled worker discovery, matching, booking, and service execution as an area for future research.

The research gap that the proposed solution addresses includes the need to develop a skilled home-service platform that integrates local provider discovery, multi-factor worker matching, transactional booking control, and automatic rematching. Connect360 utilizes the Connect360 Smart Worker Matching Algorithm (CSWMA) to rank eligible skilled service workers based on service match, trust, experience, availability, reliability, and proximity. CSWMA utilizes different worker ranking weights for Scheduled Booking and Priority Booking to ensure that the matching considers the requirements of each booking type. Connect360 also offers worker verification, booking lifecycle management, transactional worker-slot locking mechanism, worker auto-rematching for Priority Booking, and road-based tracking for active Priority Booking to enhance the service delivery process. As such, the proposed solution builds on the related literature by developing an end-to-end skilled home-service marketplace that considers worker discovery, multi-factor worker matching, transactional booking control, and automatic rematching.

III. METHODOLOGY

The methodology adopted for the Connect360 is the modular serverless one, and it incorporates the user authentication, worker verification, recommendation, booking management, AI assistance, and admin monitoring components. It is developed using three major user roles namely customer, worker, and administrator, and AWS managed services to facilitate the users with authentication, computing, storage, database, and deployment resources. The overall system design includes the React based front end, API layer, serverless back end, DynamoDB, document storage, and AI assistant.

A. System Architecture

The Connect360 application is designed based on the three-layer serverless architecture namely presentation, application, and data and intelligence services. The front end is built using the React 18 framework and the Vite tool and deployed using the Amazon S3 and the CloudFront service with HTTPS protocol. The user requests are received by the API gateway, which is protected using the AWS Cognito User Pool Authorizer. The seven AWS Lambda functions are used for the back end, which are deployed using the Python 3.11 programming language. The Lambda functions are grouped according to the application’s needs and implement the service, worker, booking, verification, administration, and other logic; the authentication and authorization are managed by the Amazon Cognito and API Gateway. The data and intelligence layer consist of the DynamoDB with single-table design and two GSIs, Amazon S3 for the document storage, and Google Gemini for the AI assistant, which are accessed using the API key stored in the AWS Secrets Manager. The resulting system architecture is shown in Fig. 1.



Fig. 1. System architecture of Connect360

This architecture reduces dependence on continuously running servers and enables independent scaling of backend functions.

B. User Authentication and Role Management

The user authentication is achieved through the AWS Cognito service, where the users can log in to the system using their email-address and password. The JWT claims are used to identify the user after the successful log in, and the backend functions accept the claims as the input to determine the action performed by the user. The API Gateway protects the API endpoints with the Cognito User Pool Authorizer, which allows authorized users only to gain access to the restricted resources. The user roles namely Customer, Worker, and Administrator have different permissions in the system, which are outlined below:

- Customer: discover workers, make bookings, monitor services, and leave reviews.
- Worker: update worker information, manage availability, view booking requests, upload verification documents.

- Administrator: control users, services, verification, revenues, and other information.

C. Worker Verification and trust model:

Workers submit their identifying, residence, and credential papers for verification. Their documents are securely stored in the private encrypted S3 bucket through pre-signed URLs. The documents are then reviewed and approved by the system administrator, and the verification results are added to the worker's trust score.

D. Intelligent Worker Matching:

Connect360 uses a server-side weighted matching engine to evaluate the service match, availability, distance, trust, experience and reliability to identify a suitable worker. Unlike a pure distance-based recommendation engine, the Connect360 algorithm is able to take multiple worker attributes into account when evaluating order assignments.

E. Connect360 Smart Worker Matching Algorithm (CSWMA)

The Connect360 Smart Worker Matching Algorithm (CSWMA) ranks eligible skilled service workers by considering multiple attributes other than the distance. For each service request, the algorithm considers service match, trust, experience, availability, reliability, and distance. Each of these attributes is normalized and combined with context-dependent weights to calculate the worker suitability score.

Let M, T, E, A, R, and D be the normalized service match, trust, experience, availability, reliability, and distance scores, respectively. All of these features are normalized to a range of $[0,1]$, where higher values indicate higher levels of influence on the worker suitability score. The individual scores represent the level of skill/service match (M), worker verification and ratings (T), experience (E), availability for the requested service (A). CSWMA defines two scoring contexts corresponding to Scheduled Booking and Priority Booking. For Scheduled Booking, the CSWMA algorithm considers the service match, trust, and experience more heavily, while also considering availability, reliability, and distance. The final score has service match, trust, experience, availability, reliability, and distance contribute 30%, 25%, 15%, 15%, 10%, and 5%, respectively.

$$SN = 0.30M + 0.25T + 0.15E + 0.15A + 0.10R + 0.05D$$

For priority booking, the algorithm favours the nearest available workers, which makes availability the most important criterion and while assigning a higher weight to availability and distance.

$$SP = 0.20M + 0.15T + 0.05E + 0.30A + 0.05R + 0.25D$$

In this case, availability accounts for 30%, distance – 25%, service match – 20%, trust – 15%, reliability – 5%, and experience – 5% of the final score. Finally, the highest-ranked worker is selected for a Priority Booking request and is eligible for automated dispatch.

F. Booking Management:

The application offers Manual Booking and Priority Booking. The latter implies that the matching engine detects an appropriate available worker and automatically presents the request to him. In case of a worker's rejection, unavailability at the moment, or timeout, the matching engine initiates rematching. The use of Atomic DynamoDB Transactions and Server-side Availability Check ensures that double booking shall not occur.

G. AI Assistant:

The application employs a basic role-aware AI assistant built with Google Gemini to provide service-specific assistance and deliver structured information, including service type, urgency, and booking details. An urgent request confirmation will trigger a Priority Booking, whereas a confirmed non-urgent request will offer worker recommendations based on similar cases. A rules-based fallback is also provided in case the AI service is unavailable.

H. Data Management and Security:

The application utilizes DynamoDB single-table with two GSIs for the core application data. User authentication and authorization are handled by Cognito JWT Claims, whereas verification documents are stored in encrypted private S3 buckets. IAM policies and Secrets Manager are used to control access to the application's secured AWS resources and AI credentials.

I. Priority Tracking and Communication:

Connect360 offers real-time tracking capabilities for Priority Booking with Leaflet.js and OpenStreetMap (OSRM) for real-road routing and ETA calculation. The calling feature allows making a call to a worker directly from an active booking using the device's default dialer, where the phone number will be offered by the calling application instead of being embedded in the Connect360 software.

J. Serverless:

The application is built with AWS's serverless technologies. The infrastructure was set up with Terraform, whereas the CloudFront service delivers the contents over HTTPS. The API Gateway serves REST API endpoints, Lambda handles the backend logic, DynamoDB stores the application's data, and S3 saves the documents.

IV. RESULTS AND ANALYSIS

Connect360 system was tested according to six experimental criteria associated with matching performance, ranking, transactional concurrency, spatial routing, AI safety, and structured AI processing. All figures are taken from actual evaluation runs and benchmarks of the system.

TABLE I
COMPARISON OF SELECTED APPROACHES AND RESEARCH GAP

Approach	Service Recommendation	Skill/Task Matching	Trust/Reliability	Spatial Allocation	Priority Booking	AI/ML Methods
Local service-provider using ML [1]	✓	✓	✓	—	—	✓
Weighted preference optimization [2]	✓	—	✓	—	—	✓
Heterogeneous graph RL for task allocation [3]	—	✓	—	✓	✓	✓
Fairness-aware task assignment [4]	✓	✓	—	✓	—	—
Human-in-the-loop task assignment [5]	—	✓	—	✓	✓	—
LLMGCL for recommendation [6]	✓	—	—	—	—	✓
Prediction-aware task assignment [8]	—	✓	—	✓	✓	—
Geographic partition framework [9]	—	✓	—	✓	—	—
Priority queue task assignment [10]	—	✓	—	✓	✓	—
Trust-aware recommender (SIoT) [11]	✓	—	✓	—	—	—
Reliable personalized recommendation [12]	✓	—	✓	—	—	✓
Skill-based job recommendation [19]	✓	✓	—	—	—	—
Cloud-native AI pipelines [20]	—	—	—	—	—	✓
Connect360 (Proposed)	✓	✓	✓	✓	✓	✓

The comparison shows that existing approaches mainly focus on individual aspects such as service recommendation, skill matching, trust, spatial allocation, or priority booking. However, these features are rarely combined within a single system. Connect360 addresses this research gap by integrating service recommendation, skill-based matching, trust and reliability, location-based allocation, priority booking, and AI/ML methods into one unified platform. This enables more personalized, reliable, and efficient service-provider matching.

A. Matching Engine Latency

Matching engine was tested according to candidate pools consisting of 10 – 5000 workers using Scheduled and Priority Booking weights. Each configuration was run 1000 times.

TABLE II
MATCHING ALGORITHM EXECUTION LATENCY

Candidates	Type	P50(ms)	P99(ms)	Mean(ms)
10	Scheduled	0.1711	0.4729	0.2106
100	Scheduled	2.5522	4.7392	2.4670
1000	Scheduled	19.2856	50.3750	22.7084
2500	Scheduled	54.9163	118.0583	60.8471
5000	Scheduled	317.5428	871.7959	308.9824
10	Priority	0.3500	1.1470	0.3770
100	Priority	2.6933	9.7360	3.1465

1000	Priority	29.1044	65.9845	30.0948
2500	Priority	72.9826	215.5545	85.2716

Matching engine showed millisecond-level performance up to 1000 candidates with average latencies of 2.467 ms for Normal and 3.147 ms for Priority Booking at 100 candidates, but significant jumps were seen past 2500 candidates which is an argument in favor of necessity to perform candidate filtering in big pools.

B. Rank Sensitivity of Dual Weighting System

The 100-worker pool was tested for rank differences when applying Normal and Priority Booking weights.

Rank similarity was evaluated using Spearman's correlation, Kendall's tau coefficient and count of rank displacement.

TABLE III
RANK SENSITIVITY OF CSWMA

Distance Tier	Spearman ρ	Kendall τ	Mean Rank Shift	Promoted	Demoted
Overall	0.8359	0.6497	13.18	50.0%	46.0%
Near (<5 km)	0.9111	0.7885	11.50	66.7%	26.7%
Mid (5–15 km)	0.8227	0.6615	12.72	55.0%	42.5%
Far (>15 km)	0.8364	0.6460	15.47	98%	70.0%

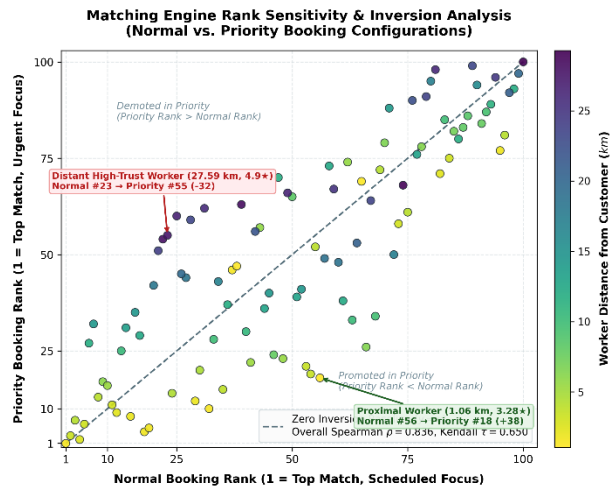


Fig. 2. Rank Sensitivity and Inversion analysis

The average rank displacement was equal to 13.18 with maximum value of 40 while closeness of workers was more often than not a reason for their prioritizing (66.7%) while the opposite was true (70.0%) for distant workers.

C. Transactional Concurrency

The booking acceptance system was tested according to number of concurrently arriving booking requests. Around 50 runs were executed for each level of concurrency.

TABLE IV
TRANSACTIONAL CONCURRENCY EVALUATION

Concurrency	Attempts	Successes	Conflicts	Conflict Rate	Double Booking
1	50	50	0	0%	0
10	500	50	450	90%	0
20	1000	50	950	95%	0
50	2500	50	2450	98%	0
100	5000	50	4950	99%	0

Across all concurrency experiments, 9,400 booking attempts were evaluated, with no double booking

observed, while 99% of booking requests were refused at a concurrency level of 100 due to transactional conflicts.

D. Spatial Routing Assessment

About 50 routes of Chennai metro city were assessed by computing the Haversine distance and OSRM road distance.

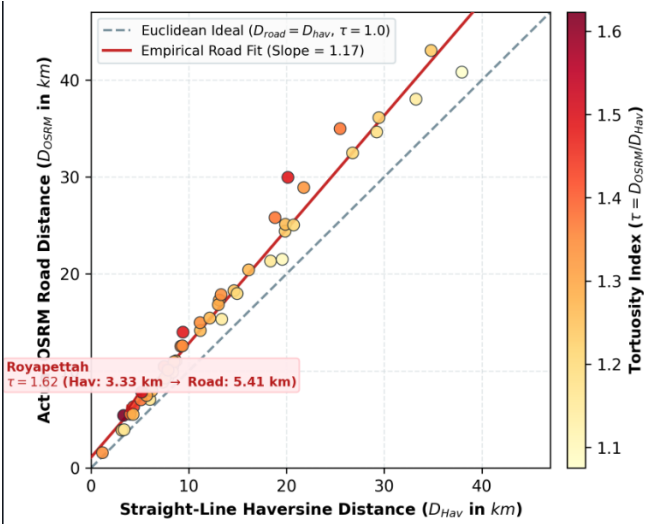


Fig. 3. Haversine vs OSRM distance by region

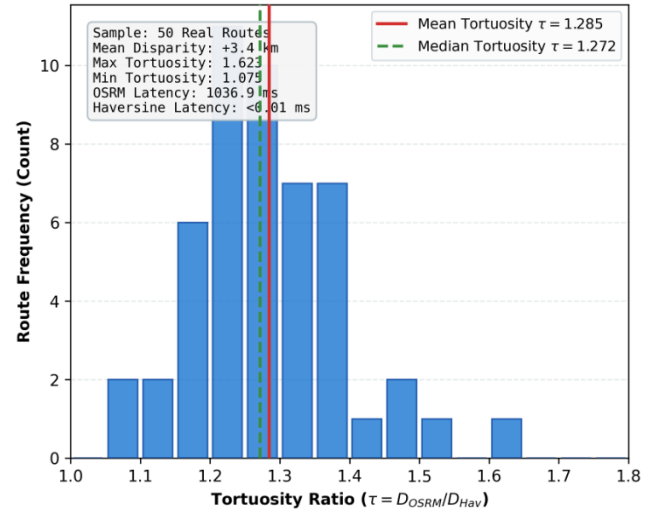


Fig. 4. Distribution of Empirical Tortuosity Index

In general, the OSRM road distance was 28.48% larger than the Haversine distance, with an average tortuosity of 1.2848. Nonetheless, the Haversine needed only 0.0296 ms, while OSRM routing required 1036.94 ms. The findings suggest that Haversine distance can be used in matching, and OSRM distance in routing.

E. AI Safety Evaluation

Redaction filter was tested on a sample of 500 test cases, 250 contained personal information, and another 250 cases were considered benign. Test cases were chosen in order to cover possible contact-information patterns, for which redaction filter had to be optimized. In terms of the present experiment, TP is the number of cases containing PII information that were successfully recognized as such, and FN stands for cases containing

PII information but recognized as benign. False positives (FP) stand for cases mistakenly recognized as containing PII information, and true negatives (TN) are the cases of personal information recognized as benign. Each test case was executed 100 times to obtain a total of 50,000 test case executions, additional tests were needed to measure computational latency of the filter.

According to the results of the experiment, there were 248 TP, 2 FN, 25 FP, and 225 TN cases, giving 90.84% precision, 99.20% recall, 94.84% F1 score, and 94.60% accuracy. There were two cases of missed personal information which involved parenthesized phone-number patterns; however, all other errors consisted of date literals in ISO format that mistakenly matched the contact-information patterns. Average execution latency was 1.919 μ s with P99 latency equal to 6.021 μ s.

TABLE V
PII REDACTION PERFORMANCE

Metric	Result
Precision	90.84%
Recall	99.20%
F1-Score	94.84%
Accuracy	94.60%
Mean Latency	1.919 μ s
P99 Latency	6.021 μ s

F. Overall Analysis

The assessment shows three main aspects of the Connect360. Firstly, the CSWMA has different rankings for workers based on the context of the booking. The difference is evident in cases where Normal and Priority booking options were used. Secondly, there is a performance factor of the Connect360 that can be described as the bigger pool of candidates resulting in higher latency of execution. Thirdly, the system was able to prevent double bookings from happening despite the intense use of the system.

The routing analysis and the AI assistant evaluation can be seen as separate validation tests for the Priority Booking feature and conversational-type assistance, respectively. Both were used to demonstrate the actual execution of the Connect360 features without engaging in performance claims that could be invalid for the customer.

V. DISCUSSION

From the evaluation results, it is clear that the CSWMA is capable of changing the ranking of the worker depending on the context of the booking process due to different weightings in the case of the Scheduled and Priority Booking.

In terms of the latency evaluation, it is obvious that the complexity of the worker ranking procedure increases linearly with respect to the size of the candidate set. This means that the optimization of the candidate set becomes crucial in terms of high concurrency.

Based on the results of the concurrency evaluation, it is possible to say that the transactional booking procedure is enough to prevent the double-booking. This means that there were no examples of double-booking in all concurrency scenarios that were evaluated.

The evaluation results in terms of the spatial routing demonstrate that the road-based distance calculation is more accurate than the straight line distance since the roads are the main means of transportation for the majority of the workers. What is more, it is important to note that the distance calculation is very important for the accuracy of the Priority Booking ranking because the possibility of the arrival of the worker to the booking location depends on their transportation possibilities.

In general, according to the results of the AI evaluation, it is possible to say that the system is capable of handling the service-specific interactions and ensuring the privacy of the contact information of the workers. First of all, in terms of the PII redaction evaluation, the high recall of the system was observed along with some space for the reduction of the number of false positives. Besides, the evaluation of the structured output demonstrated the adequacy of the service, urgency level, confirmation instruction, and structured response of the system.

VI. CONCLUSION

Connect360 is a hypothetical serverless skilled home-service platform that helps its customers to find, select, and book certified service workers. The Connect360 Smart Worker Matching Algorithm for Scheduled and Priority Booking is applied to the platform in accordance with the different scoring settings in the specific context of requested services.

As can be seen from the proof-of-concept prototype implementation, the possibility of using the service-worker matching, transactional booking, worker verification, road-based routing, AI-driven service interaction, and serverless computing on a single home-service platform exists. The findings of this research have proved the possibility to effectively rank workers by different criteria, ensure booking consistency when there is a great number of service requests, and make adequate service allocation and interaction secure enough.

The next steps in the development of the implemented solution include optimization of the candidate selection settings in case of a large number of service workers, minimization of the routing delay, setting up of the AI

parameters in ambiguous cases of service contexts, and using of the prototype in other applications.

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